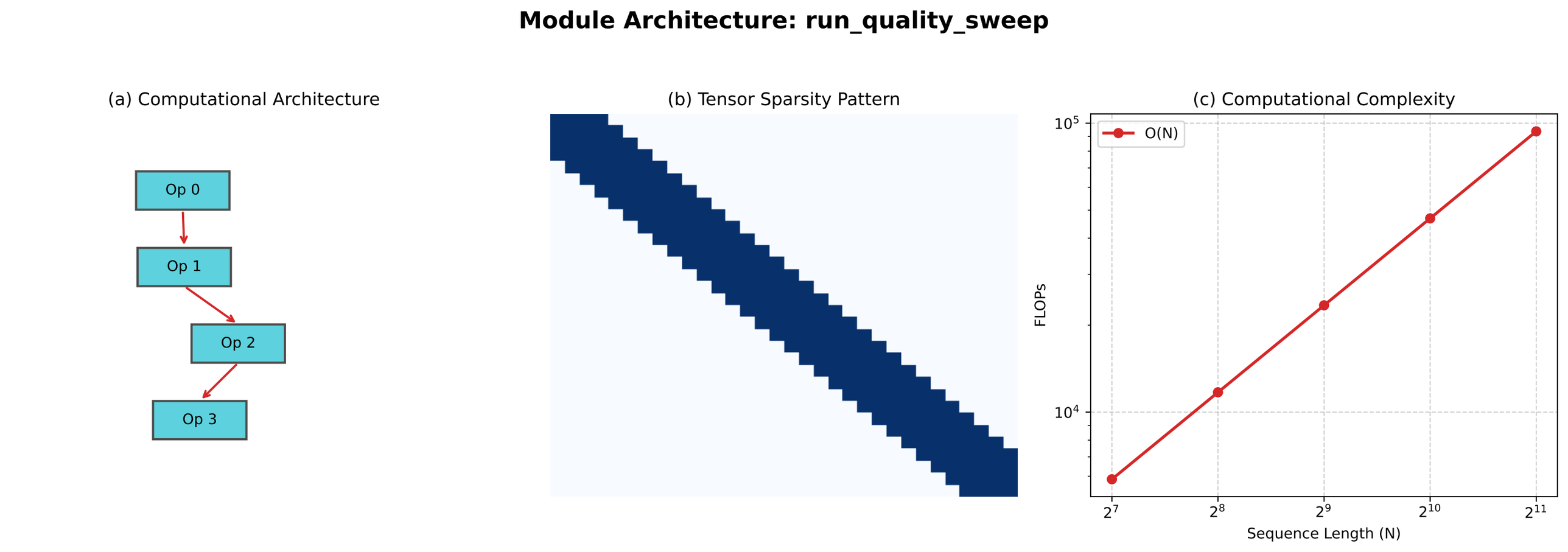Module Architecture: run_quality_sweep